

Lab Experiment: Dynamic Memory Allocation in OOP (C++)

Aim:

To write a C++ program that demonstrates dynamic memory allocation using classes and objects.

Theory:

Dynamic memory allocation allows memory to be allocated at runtime using operators new and delete.

In Object-Oriented Programming, we can allocate memory for objects dynamically, which helps in efficient memory management, especially when the size of data is unknown at compile time.

- new operator → allocates memory dynamically.
- delete operator → frees the dynamically allocated memory.

Program (C++):

```
#include <iostream>
using namespace std;

class Student {
    string name;
    int *marks;

public:
    Student(string n, int m) {
        name = n;
        marks = new int;
        *marks = m;
    }

    void display() {
        cout << "Student Name: " << name << endl;
        cout << "Marks: " << *marks << endl;
    }

    ~Student() {
        delete marks;
        cout << "Memory released for " << name << endl;
    }
};

int main() {
    Student *s1 = new Student("Alice", 85);
    Student *s2 = new Student("Bob", 90);

    cout << "Displaying Student Details:\n";
    s1->display();
    s2->display();

    delete s1;
    delete s2;

    return 0;
}
```

Expected Output:

```
Displaying Student Details:  
Student Name: Alice  
Marks: 85  
Student Name: Bob  
Marks: 90  
Memory released for Alice  
Memory released for Bob
```

Viva Questions & Answers:

Q1. What is dynamic memory allocation in C++?

■ It is the process of allocating memory at runtime using operators like new and delete.

Q2. Why do we use new instead of malloc() in C++?

■ new initializes the object and calls the constructor, while malloc() only allocates raw memory and does not call constructors.

Q3. What is the difference between delete and delete[]?

■ delete is used to free memory of a single object, while delete[] is used to free memory allocated for an array of objects.

Q4. What is the role of a destructor in dynamic memory allocation?

■ A destructor is used to release dynamically allocated memory and clean up resources before an object is destroyed.

Q5. What happens if we don't free dynamically allocated memory?

■ It leads to a memory leak, where allocated memory remains unusable until the program terminates.